

AMC 10 Crash Course • Block 3 • Number Theory

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

Lesson 3.1. Divisibility, Prime Factorization, Number and Sum of Divisors

THEORY

Divisibility rules. For 3 and 9 — the digit sum; for 11 — the alternating digit sum (right to left, alternating signs); for 4 — the last two digits, for 8 — the last three. Composite rules are built from coprime parts: divisibility by 12 means divisibility by 3 **and** by 4 at once; checking only one part is a classic mistake.

Prime factorization is a number's passport: $720 = 2^4 \cdot 3^2 \cdot 5$. Nearly every question about divisors, GCD, LCM, or powers is read straight off the exponents. In any problem about a specific number, factoring it is reflex number one.

If $n = p^a q^b r^c \dots$, then the number of divisors $= (a+1)(b+1)(c+1)\dots$, and the sum of divisors $= (1 + p + \dots + p^a)(1 + q + \dots + q^b)\dots$, each bracket equal to $\frac{p^{a+1} - 1}{p - 1}$.

Number of divisors. A divisor is a choice of exponent for each prime: from 0 to a , that is $a+1$ options, and the options multiply. Conditional counts work the same way: perfect-square divisors are choices of even exponents; divisors that are multiples of 6 keep the exponents of 2 and 3 at least 1.

Sum of divisors — the same brackets, but with sums of powers instead of counts. A useful check: the number of divisors is odd only for perfect squares — divisors pair up as d and n/d , and only a square has the unpaired divisor $\sqrt{n}$.

WORKED EXAMPLES

EXAMPLE 1 • DIVISIBILITY RULES

For which digit a is the number $4a16$ divisible by 36?

→ $36 = 4 \cdot 9$, the parts are coprime — check both. By 4: the last two digits are 16, divisible for every a . By 9: the digit sum $4 + a + 1 + 6 = 11 + a$ must be divisible by 9, so $a = 7$. Check: $4716 / 36 = 131$. The trap is checking only the 9 and forgetting the 4 (free here, but you must check it).

EXAMPLE 2 • NUMBER OF DIVISORS

How many positive divisors does 720 have?

→ $720 = 2^4 \cdot 3^2 \cdot 5$. Number of divisors: $(4+1)(2+1)(1+1) = 5 \cdot 3 \cdot 2 = \mathbf{30}$. The trap is multiplying the exponents themselves ($4 \cdot 2 \cdot 1 = 8$): forgetting the “+1” is the most common error with this formula. Exponent 0 is a choice too.

EXAMPLE 3 • SUM OF DIVISORS

Find the sum of all positive divisors of 200.

→ $200 = 2^3 \cdot 5^2$. Sum $= (1 + 2 + 4 + 8)(1 + 5 + 25) = 15 \cdot 31 = \mathbf{465}$. The brackets multiply because every divisor is exactly one product “power of 2 times power of 5”, and expanding the brackets lists each of them exactly once.

EXAMPLE 4 • DIVISORS WITH A CONDITION

How many divisors of 360 are multiples of 6?

→ $360 = 2^3 \cdot 3^2 \cdot 5$. A divisor is a multiple of 6 when the exponent of 2 is 1, 2, or 3 (three options), of 3 — 1 or 2 (two), of 5 — 0 or 1 (two): $3 \cdot 2 \cdot 2 = \mathbf{12}$. Equivalent move: multiples of 6 among the divisors are $6d$ where d runs over divisors of $360/6 = 60$, and there are $(2+1)(1+1)(1+1) = 12$ of those.

INDEPENDENT SET

1. For which digit a is the number $73a5$ divisible by 9?
2. For which digit a is the number $6a352$ divisible by 11?
3. How many positive divisors does 200 have?
4. Find the sum of all positive divisors of 72.
5. How many even divisors does $2^5 \cdot 3^4$ have?
6. Find the least positive integer with exactly 10 divisors.
7. How many divisors of 3600 are perfect squares?
8. How many integers from 1 to 60 have an odd number of divisors?
9. ★ How many divisors of 1,000,000 are neither perfect squares nor perfect cubes?

Lesson 3.2. GCD and LCM, the Euclidean Algorithm, Coprimality

THEORY

Via factorization. The GCD takes the **smaller** exponent of each prime, the LCM the **larger**: for $2^3 \cdot 3^2 \cdot 5$ and $2^2 \cdot 3^3 \cdot 7$, $\text{GCD} = 2^2 \cdot 3^2 = 36$ and $\text{LCM} = 2^3 \cdot 3^3 \cdot 5 \cdot 7 = 7560$. Since $\min + \max = \text{sum of the exponents}$, we get the key identity below.

$$\text{gcd}(a, b) \cdot \text{lcm}(a, b) = a \cdot b. \quad \text{Euclid: } \text{gcd}(a, b) = \text{gcd}(b, a \bmod b).$$

The Euclidean algorithm — for when factoring is slow: replace the larger number by its remainder upon division by the smaller, and repeat; the last nonzero remainder is the GCD. It works with letters too: $\text{gcd}(a, b) = \text{gcd}(b, a - kb)$ for any integer k — that is how you tame gcds of expressions like $6n + 5$ and $3n + 2$.

Coprimality: $\text{gcd} = 1$, no common primes. Consecutive integers n and $n + 1$ are always coprime. The standard substitution: $a = gx$, $b = gy$, where g is the gcd and $\text{gcd}(x, y) = 1$; then $\text{lcm} = gxy$.

Word-problem dictionary. “Cut into equal largest pieces”, “share evenly with nothing left” — GCD. “Meet again, flash together, coincide”, “the schedules line up” — LCM. Mixing them up is trap number one in word problems; the GCD is at most each number, the LCM at least.

WORKED EXAMPLES

EXAMPLE 1 • EUCLIDEAN ALGORITHM

Find $\text{gcd}(1071, 462)$.

→ $1071 = 2 \cdot 462 + 147$; then $462 = 3 \cdot 147 + 21$; then $147 = 7 \cdot 21 + 0$. The last nonzero remainder: $\text{gcd} = \mathbf{21}$. Three lines instead of factoring — on large numbers Euclid is always faster.

EXAMPLE 2 • GCD • LCM = AB

Two numbers have gcd 6 and lcm 180; one of the numbers is 30. Find the other.

→ The product of the numbers equals $\text{gcd} \cdot \text{lcm} = 6 \cdot 180 = 1080$. The other number: $1080 / 30 = \mathbf{36}$. Check: $\text{gcd}(30, 36) = 6$, $\text{lcm} = 180$. The identity removes all guesswork.

EXAMPLE 3 • VIA EXPONENTS

Find the gcd and lcm of $2^3 \cdot 3^2 \cdot 5$ and $2^2 \cdot 3^3 \cdot 7$.

→ For each prime: minimum into the gcd, maximum into the lcm. $\text{Gcd} = 2^2 \cdot 3^2 = \mathbf{36}$; $\text{lcm} = 2^3 \cdot 3^3 \cdot 5 \cdot 7 = \mathbf{7560}$. The primes 5 and 7 enter the lcm but not the gcd: the minimum of their exponents is zero. Control: $36 \cdot 7560 = 360 \cdot 756$, the product of the numbers.

EXAMPLE 4 • WORD PROBLEM

Two ropes, 84 m and 60 m, are cut into equal pieces of the greatest possible length. How many pieces result?

→ “Equal largest pieces” means $\text{gcd}(84, 60) = 12$ m. Pieces: $84/12 + 60/12 = 7 + 5 = \mathbf{12}$. The trap is grabbing the lcm: you cannot cut 420-meter pieces from these ropes. $\text{Gcd} \leq$ each length — a quick sanity check.

INDEPENDENT SET

1. Find $\text{gcd}(48, 180)$.
2. Find $\text{lcm}(24, 36)$.
3. Use the Euclidean algorithm to find $\text{gcd}(391, 323)$.
4. Two numbers have gcd 8 and product 1920. Find their lcm.
5. Two buses leave a station every 12 and every 18 minutes; at noon they left together. When do they next leave at the same time?
6. How many integers from 1 to 100 are coprime with 100?
7. Find the gcd of $6n + 5$ and $3n + 2$ (for an arbitrary positive integer n).
8. How many ordered pairs (a, b) of positive integers have $\text{gcd}(a, b) = 5$ and $\text{lcm}(a, b) = 60$?
9. ★ Find the sum of all positive integers $n \leq 60$ with $\text{gcd}(n, 18) = 6$.

Lesson 3.3. Remainders: Mod Arithmetic, Last Digits, Power Cycles

THEORY

Mod arithmetic. Writing $a \equiv b \pmod{n}$ means: a and b leave the same remainder upon division by n .

Remainders may be added, subtracted, and multiplied BEFORE computing: the remainder of a product is the remainder of the product of remainders. Replace a giant expression by small remainders first, then compute. Useful: a number's remainder mod 9 equals the remainder of its digit sum; mod 10 — its last digit.

The last digit is the remainder mod 10. Last digits of powers run in cycles: for 7 it is 7, 9, 3, 1 with period 4; for 2 — 2, 4, 8, 6; for 3 — 3, 9, 7, 1. Divide the exponent by the cycle length and take the remainder. The trap: remainder 0 means the **last** position of the cycle, not the first.

Remainder of a power: 1) list the remainders $a^1, a^2, a^3, \dots \pmod{n}$ until they repeat; 2) exponent mod cycle length; 3) remainder 0 → the last term of the cycle.

The last two digits are the remainder mod 100. Cycles are longer, but the tools are the same; memorize the binomial shortcut: $11^n = (10 + 1)^n \equiv 10n + 1 \pmod{100}$ — every term containing 100 vanishes.

Powers mod n cycle too: 2 mod 7 cycles through 2, 4, 1 with length 3; for 3 mod 8, already $3^2 \equiv 1$, so every even power of 3 gives 1. Once you find a power with remainder 1, everything after collapses instantly. This is the main tool for “find the remainder of 2^{2026} ” problems.

WORKED EXAMPLES

EXAMPLE 1 • LAST DIGIT

Find the last digit of 7^{2026} .

→ The cycle of 7: 7, 9, 3, 1, length 4. Exponent: $2026 = 4 \cdot 506 + 2$, remainder 2 — the second position of the cycle: **9**. The trap: students forget to reduce the exponent mod 4 and grab something from “2026” itself — the cycle is driven by the exponent, not the base.

EXAMPLE 2 • A POWER WITH REMAINDER 1

Find the remainder when 3^{100} is divided by 8.

→ $3^2 = 9 \equiv 1 \pmod{8}$. Then $3^{100} = (3^2)^{50} \equiv 1^{50} = \mathbf{1}$. One line: once some power gives remainder 1, every multiple of that exponent gives 1 too. No need to chase the cycle further.

EXAMPLE 3 • LAST TWO DIGITS

Find the last two digits of 7^{2024} .

→ Mod 100: $7^2 = 49$, $7^3 = 343 \equiv 43$, $7^4 = 2401 \equiv 1$. The cycle has length 4, and 2024 is divisible by 4 — remainder 0, i.e. the LAST position of the cycle: $7^{2024} \equiv 1 \pmod{100}$, last two digits **01**. The classic trap: reading remainder 0 as the first position and answering 07.

EXAMPLE 4 • SUBSTITUTING A REMAINDER

A number N leaves remainder 3 upon division by 7. What remainder does $N^2 + 2N$ leave?

→ Substitute the remainder for the number: $N \equiv 3 \pmod{7}$, so $N^2 + 2N \equiv 9 + 6 = 15 \equiv \mathbf{1} \pmod{7}$. We never need N itself — remainders add and multiply in place of numbers. The intermediate 15 exceeds the modulus — remember to reduce it at the end.

INDEPENDENT SET

1. Find the remainder when 123,456 is divided by 9.
2. Find the last digit of 3^{2027} .
3. Find the remainder when 2^{100} is divided by 7.
4. A number satisfies $n \equiv 5 \pmod{6}$. Find the remainder of $4n + 3$ upon division by 6.
5. Find the last digit of $2^{2026} + 3^{2026}$.
6. Find the remainder when $1 + 2 + 3 + \dots + 100$ is divided by 9.
7. Find the last two digits of 11^{2025} .
8. Find the remainder when 2^{2026} is divided by 13.
9. ★ Find the remainder when $2^{2026} + 2026^2$ is divided by 7.

Lesson 3.4. Factorials and Legendre's Formula, Digit Problems, Number Bases

THEORY

Legendre's formula. The exponent of a prime p in $n!$ is a sum of floors: how many numbers up to n are divisible by p , by p^2 , by p^3 , ... Each next term counts the extra factor of p contributed by multiples of the higher power. The trap is stopping after the first term: 25s in $100!$ contribute not 20 but $20 + 4$.

Exponent of p in $n! = \frac{n}{p} + \frac{n}{p^2} + \frac{n}{p^3} + \dots$ (each term is a floor). Trailing zeros of $n! =$ the exponent of 5.

Trailing zeros. A zero is born from a pair $2 \cdot 5$, and factorials always contain more 2s than 5s — so count only the 5s. For $100!$: $20 + 4 = 24$ zeros. The inverse problem is treacherous: the zero count jumps by 2 at multiples of 25, so some values (exactly 5 zeros, for instance) are never attained.

Digit problems. A two-digit number is $10a + b$, not “ ab ”. Key identities: $N + \text{reversal} = 11(a + b)$; $N - \text{reversal} = 9(a - b)$. Once the number is written through its digits, the problem becomes an equation.

Number bases — the minimum. The string 132_5 means $1 \cdot 25 + 3 \cdot 5 + 2 = 42$. To decimal — multiply out the place values; from decimal — divide by the base and collect remainders bottom-up. Digits must be smaller than the base: “digit 7 in base 5” signals an error.

WORKED EXAMPLES

EXAMPLE 1 • TRAILING ZEROS

How many trailing zeros does $100!$ have?

→ Count the 5s: $\lfloor 100/5 \rfloor = 20$, $\lfloor 100/25 \rfloor = 4$, $\lfloor 100/125 \rfloor = 0$. Total $20 + 4 = \mathbf{24}$. The trap is answering 20, forgetting that 25, 50, 75, and 100 each carry a SECOND five — exactly what the term $\lfloor 100/25 \rfloor$ adds.

EXAMPLE 2 • LEGENDRE’S FORMULA

Find the exponent of 3 in the prime factorization of $30!$.

→ $\lfloor 30/3 \rfloor + \lfloor 30/9 \rfloor + \lfloor 30/27 \rfloor = 10 + 3 + 1 = \mathbf{14}$. The next term $\lfloor 30/81 \rfloor = 0$ — the series stops by itself. Meaning of the terms: 10 multiples of 3, of which 3 are multiples of 9 (a second three), one is a multiple of 27 (a third).

EXAMPLE 3 • DIGITS AND REVERSAL

A two-digit number is added to its reversal, giving 132. What is the sum of the number’s digits?

→ $(10a + b) + (10b + a) = 11(a + b) = 132$, so $a + b = \mathbf{12}$. The number itself is not determined (48, 57, 66, 75, 84, 93 all work) — but that is not the question. Do not chase extras: the identity hands over the answer in one line.

EXAMPLE 4 • BASE CONVERSION

Write 2026 in base 5.

→ Divide by 5, remainders bottom-up: $2026 = 5 \cdot 405 + 1$; $405 = 5 \cdot 81 + 0$; $81 = 5 \cdot 16 + 1$; $16 = 5 \cdot 3 + 1$; $3 = 5 \cdot 0 + 3$. Read the remainders backwards: $\mathbf{31101_5}$. Check by place values: $3 \cdot 625 + 1 \cdot 125 + 1 \cdot 25 + 0 + 1 = 2026$. Correct.

INDEPENDENT SET

- How many trailing zeros does $25!$ have?
- Find the exponent of 2 in the factorization of $20!$.
- Convert 110110_2 to decimal.
- How many trailing zeros does $130!$ have?
- How many two-digit numbers exceed their reversal by exactly 27?
- Find the largest k for which 12^k divides $30!$.
- In which base does the string 132 represent the decimal number 42?
- Find the least n for which $n!$ ends in exactly 10 zeros.
- ★ Find the least positive integer m such that NO factorial $n!$ ends in exactly m zeros.

Solutions to the independent sets

LESSON 3.1

1. $7 + 3 + 5 = 15$; we need $15 + a \equiv 0 \pmod{9}$, so $a = 3$ (sum 18).
2. $2 - 5 + 3 - a + 6 = 6 - a$ must be divisible by 11: $a = 6$. Check: $66,352 = 11 \cdot 6032$.
3. $(3+1)(2+1) = 12$.
4. $(1 + 2 + 4 + 8)(1 + 3 + 9) = 15 \cdot 13 = 195$.
5. Total divisors: $6 \cdot 5 = 30$; odd ones (exponent of 2 equal to 0): 5. Even: $30 - 5 = 25$.
6. Candidates: $2^9 = 512$ and $2^4 \cdot 3 = 48$. Put the large exponents on the small primes: the answer is **48**.
7. Even exponents: for $2 - 0, 2, 4$ (three options), for 3 and for $5 - 0$ or 2 (two each): $3 \cdot 2 \cdot 2 = 12$.
8. Only perfect squares have an odd number of divisors. From 1 to 60: 1, 4, 9, 16, 25, 36, 49 — **7 numbers**.
★ Total divisors: $7 \cdot 7 = 49$. Squares (even exponents 0, 2, 4, 6): $4 \cdot 4 = 16$. Cubes (exponents 0, 3, 6): $3 \cdot 3 = 9$. Sixth powers (0, 6): $2 \cdot 2 = 4$. Answer: $49 - 16 - 9 + 4 = 28$.

LESSON 3.2

1. $2^2 \cdot 3 = 12$.
2. $\text{Gcd} = 12$, so $\text{lcm} = 24 \cdot 36 / 12 = 72$.
3. $323 = 4 \cdot 68 + 51$; $68 = 51 + 17$; $51 = 3 \cdot 17$. $\text{Gcd} = 17$ (indeed $391 = 17 \cdot 23$, $323 = 17 \cdot 19$).
4. $\text{Lcm} = 1920 / 8 = 240$.
5. $\text{Lcm}(12, 18) = 36$: at **12:36**.
6. $100 - 50$ (even) $- 20$ (multiples of 5) $+ 10$ (multiples of 10, subtracted twice) = **40**.
7. $(6n + 5) - 2(3n + 2) = 1$, so the gcd divides 1: it equals **1** for every n .
8. By coprimality each prime power (4 and 3) goes wholly to x or to y : $2 \cdot 2 = 4$ pairs. They are (5, 60), (60, 5), (15, 20), (20, 15).
★ $\text{Gcd}(n, 18) = 6 = 2 \cdot 3$ requires: n even and divisible by 3 exactly once. Multiples of 6 up to 60: ten of them; strike out the multiples of 18 (18, 36, 54). Left: 6, 12, 24, 30, 42, 48, 60; the sum is **222**.

LESSON 3.3

1. Digit sum $1+2+3+4+5+6 = 21 \equiv 3 \pmod{9}$.
2. $2027 \equiv 3 \pmod{4}$ — the third position of the cycle: 7.
3. Cycle length 3; $100 \equiv 1 \pmod{3}$ — the first position: **2**.
4. $4 \cdot 5 + 3 = 23 \equiv 5 \pmod{6}$.
5. $2026 \equiv 2 \pmod{4}$: 2^{2026} ends in 4, 3^{2026} in 9. Sum 13, last digit **3**.
6. Sum = $100 \cdot 101/2 = 5050$; digit sum $10 \equiv 1 \pmod{9}$.
7. $10 \cdot 2025 + 1 = 20,251 \equiv 51 \pmod{100}$.
8. $2^{12} \equiv 1 \pmod{13}$; $2026 = 12 \cdot 168 + 10$, so $2^{2026} \equiv 2^{10} = 1024 = 78 \cdot 13 + 10 \equiv 10$.
★ Powers of 2 mod 7: 2, 4, 1 (cycle 3); $2026 \equiv 1 \pmod{3}$, so $2^{2026} \equiv 2$. Next, $2026 = 7 \cdot 289 + 3$, so $2026 \equiv 3$ and $2026^2 \equiv 9 \equiv 2$. Sum: $2 + 2 = 4$. The trap: the exponent is reduced by the cycle length (mod 3), the base mod 7; mixing these up is the standard error.

LESSON 3.4

1. $5 + 1 = 6$.

2. $10 + 5 + 2 + 1 = 18$.

3. $32 + 16 + 4 + 2 = 54$.

4. $\lfloor 130/5 \rfloor + \lfloor 130/25 \rfloor + \lfloor 130/125 \rfloor = 26 + 5 + 1 = 32$.

5. $9(a - b) = 27$, so $a - b = 3$: the numbers 30, 41, 52, 63, 74, 85, 96 — 7 in all.

6. 2s: $15 + 7 + 3 + 1 = 26$; 3s: $10 + 3 + 1 = 14$. We need $2k \leq 26$ and $k \leq 14$: $k = 13$. The bottleneck is the 2s, despite there being “more” of them: each 12 spends two.

7. $b^2 + 3b - 40 = 0$, roots 5 and -8 : base 5. The digits 1, 3, 2 are below five — the string is legal.

8. For $44!$: $\lfloor 44/5 \rfloor + \lfloor 44/25 \rfloor = 8 + 1 = 9$ zeros; for $45!$: $9 + 1 = 10$. The answer is 45. The zero count changes only when crossing a multiple of five — nothing to check in between.

★ At each multiple of five the zero counter increases by that number’s exponent of 5. For $24!$: $\lfloor 24/5 \rfloor = 4$ zeros; 25 brings TWO fives at once, and $25!$ already has 6 zeros. The value 5 is jumped over and never attained: $m = 5$. Further skips occur at every multiple of 25 (for instance 11, between $49!$ and $50!$).